Spatial–Spectral Local Domain Adaption for Cross Domain Few Shot Hyperspectral Images Classification

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***Abstract*— The traditional methods of hyperspectral image (HSI) classification are based on the sufficient labeled data. In real life, we often encounter that the target domain corresponding to the classification task has only a small amount of labeled data, but the source domain has enough labeled data. However, the distribution of the source domain is different from the distribution of the target domain. Thus, the labeled data of the source domain cannot be applied to the target domain directly. This article proposes a new method to solve the cross-domain few shot problem of HSI classification. In the proposed method, the local spatial alignment and the spectral alignment are simultaneously introduced to transfer the knowledge from the source domain to the target domain. Besides, to extract the domain-specific features, we balance the domain-invariant features and the domain-specific features by a weakly parameter-shared mechanism. The two modules together can narrow the distance between two domains and make the model perform well on the target domain. Experiments conducted on four different target domain datasets demonstrate the effectiveness of our method.**

***Index Terms*— Cross domain few shot learning (CDFSL), hyperspectral image (HSI), local alignment, weakly parameter-shared mechanism.**

I. Introduction

HYPERSPECTRAL images (HSIs) are acquired in many continuous spectral bands of the entire spectrum [1]. It has a wide range of applications in various fields, such as land change detection, classification, identification, detection, and so on. Since HSIs have a large number of bands, classification becomes one of its main challenges.

According to whether spatial information is used or not, early HSI classification methods can be roughly divided into two categories. One category only relies on spectral information, such as the distance and similarity between spectral features as the classification basis [2], [3]. The other utilizes both the spatial information and the spectral information to realize the HSI classification. Based on the pixel-level classification of spectral information, Markov random field (MRF) [4] is introduced to refine the classification results by spatial information, and [5] proposed a model that uses a 3-D filter to directly extract the spatial–spectral feature. Unlike ordinary images, the special imaging method of HSI results in dissimilar spectra of the same class of targets in different positions, but similar spectra of different classes of targets in neighboring positions. Therefore, the classification effect of the method based on the spectral information is not as good as that of joint spatial-spectrum features.

Since 2012, deep learning has developed rapidly in various fields. Therefore, researchers have tried to solve the problem of HSI classification by deep learning and achieved significant results. Deep learning methods mainly rely on joint spectral-spatial features. In 2013, Lin *et al.* [6] used the deep learning methods to solve the problem of HSI classification for the first time. They first performed principal component analysis (PCA) on the HSI, reducing the number of channels from hundreds to tens, and extracted the neighborhood features of the pixels to be classified into sparse autoencoder (SAE) to obtain the spectral-spatial features. On this basis, Chen *et al.* [7], [8] proposed a dual-channel structure, introducing a spectral dimensional channel that uses spectral features as input, and used SAE and deep belief network (DBN) to extract features, respectively. It realizes the feature fusion with the output of the spectral-spatial channel. At the same time, researchers try to apply convolutional neural network (CNN) to HSI classification. 1D-CNN is used to extract spectral features [9], [10], spatial features [11], [12] are extracted by 2D-CNN and 3D-CNN is used to extract spectral-spatial joint features [13]. Different from the traditional methods, the deep learning methods do not rely on manually design features and are more robust. However, general deep learning methods rely on sufficient labeled data. The cost of data labeling is huge and time consuming. In reality, we often cannot obtain sufficient data and labels.

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To assist classification with unlabeled data, researchers turned their attention to semisupervised (SS) learning. Liu *et al.* [14] proposed a SS CNN that can automatically learn features from HSIs, which jointly considers supervised and unsupervised loss for training. In addition, active learning [15], [16], [17] is also used to solve the problem of lacking samples. To come up with the problem of insufficient labeled samples, researchers proposed a lot of methods, but these methods are all implemented under the premise that the distributions of the training dataset and the testing dataset are consistent. In the problem of realizing the hyperspectral data classification, we often find that the target datasets lack labels, and at the same time, there are some datasets with sufficient labels. We take datasets with a large number of labels as the source domain and datasets with a lack of labels as the target domain. To solve the problem of existing distribution discrepancy, a series of cross-domain learning methods have been proposed by researchers; the methods can be divided into three parts roughly, methods based on distribution alignment [18], [19] [20], methods based on feature selection [21], [22] [23], and methods based on subspace learning [24], [25] [26].

Methods based on distribution alignment try to build a model which minimizes the distance between the distribution of the source domain and the target domain. Xia *et al.* [19] viewed the domain adaptation problem as an ensemble strategy and proposed a method named transfer component analysis (TCA) to fuse the feature space extracted by the source and target domains. Methods based on feature selection assume that the two domains share a common feature space. The goal of these methods is to extract the shared features of two domains and build a deep learning model based on the extracted features. Li *et al.* [23] integrated structure preservation and feature selection into a unified problem by the method named joint feature selection and structure preservation (FSSP). Methods based on subspace learning transform the target domain and the source domain into a shared subspace where two domains have similar distributions. Zhu *et al.* [27] proposed a network structure named deep subdomain adaptive network (DSAN), the network can realize the alignment of related subdomains through local maximum mean discrepancy (LMMD); Qu textitet al. [28] proposed a physical constraint transfer learning method, which transformed the data to a shared space to reduce the domain discrepancy.

The above is the single-source domain problem. In addition, in the face of problems with multiple source domains, scholars have also proposed many different methods. Mansour *et al.* [29] considered that the distribution of the target domain can be represented by a mixture of multiple source domains, and introduced the weighted classifiers of the source domain into the model. Zhao *et al.* [30] considered not only the distance between the source domain and the target, but also the similarity between the source domain samples and the target domain samples. Liang *et al.* [31] proposed a multisource attention fusion method named AMF-few shot learning (FSL), the method utilizes an attention mechanism to achieve multisource fusion and transfer the knowledge learned from different source domains to the target domain.

These methods are only applicable to the case that the source and target domains belong to the same category space, so the classification of the unique category in the target domain cannot be achieved.

To solve the problem that the category space of two domains is different, Mei *et al.* [32] proposed a companion feature learning network, which first trains a deep learning model based on the source domain, and then, they use the fully connected (FC) feature layer of the classification model to construct the network. Yang *et al.* [33] constructed a dual-branch model to obtain the spatial–spectral features, they also introduced a fine-tuning mechanism for limited samples at the same time. So the model can also have a better classification effect on the unique category of the target domain. These methods implement transfer learning under the inconsistent category spaces of the two domains, but the fine-tuning mechanism requires there is only a small difference between the distribution of the source domain and the target domain. In the case of obvious differences between the two domains, it cannot play a good role. Thus, researchers pay more attention to how to utilize the limited labeled samples of the target domain to get a better classification effect. Li *et al.* [34] constructed a deep cross-domain few shot learning network, named DCFSL, which introduces the domain adaptive methods into few shot learning. The network utilizes a global alignment module to minimize the distribution distance between two domains. Although this method improves the classification accuracy of the model in the target domain to a certain extent, it still has the following limitations.

1) Different domains belong to different distributions, but DCFSL uses the same feature extraction network for the two domains. The model can only retain some invariant information in the two domains (domain-invariant information), and cannot retain the domain-specific information. Some information of categories in the target domain may not have representation in the source domain, it is not enough to classify samples only by the domain-invariant information.
2) The model uses the global domain discriminator to realize the alignment of different data distributions. The global alignment will make two domains align well on the global marginal distribution, but it fails to align the categories within the domain. The features differences are not considered, which will cause the features mapped to the wrong category.
3) The method of the cross-domain HSI classification is proposed to solve the problem that the source domain has sufficient labeled data, but the target domain has only a little labeled samples. The few shot learning method adopted by the model only uses the labeled samples instead of the whole samples. There is a lot of information in the unlabeled samples which are ignored during the training process. Because of this, the features extracted by the model do not represent the whole dataset.

To solve the above problems, a spatial–spectral local domain adaption network is proposed. First, the source domain and

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the target domain are globally dimensional reduced. The weakly shared mechanism is used to retain the domain-specific features. Then, the global alignment and the local alignment are performed on the two domains at the same time. It realizes the alignment of the local features based on the alignment of the global representation of the two datasets. To make full use of the spectral information of HSI, a spatial–spectral local alignment module is proposed. The main contributions of this work are as follows.

1) To extract both the domain-invariant features and the domain-specific features for the HSI classification, a weakly parameter shared mechanism is introduced into the proposed network. For different domains, the feature extraction networks in the source domain and the target domain have the same the network structure, but the parameters are different. A constraint is added to make sure that the network can obtain the domain-specific features and the domain-invariant features simultaneously.
2) Different from the global alignment, the local alignment pays more attention to the differences between local feature distributions. The local alignment module is composed of local space alignment and spectral alignment, which can align the local feature distributions of two domains from both spatial and spectral perspectives.
3) PCA is used to reduce the global dimensions of the source domain and the target domain, which makes full use of the unlabeled sample information. Thus, our method can select the most important spectral dimension from an overall perspective, and the features extracted by the feature extraction network will be more representative.

The rest of this article is organized as follows. Section II introduces the latest development of domain adaptation in HSI classification. Section III introduces the specific implementation of the proposed spatial–spectral local domain adaption network, and Section IV gives the experimental results of this method on four different target datasets. Conclusions are given in Section V.

II. Related Work

A. HSIs Classification

The development of HSI classification based on deep learning can be divided into three stages. In the first phase, researchers have focused on 1D-CNN and 2D-CNN since 2015. The methods of HSI classification use 1D-CNN to extract features along the spectral dimension of the image. Mei *et al.* performed 1D-CNN on the image to extract deep spectral features, [35] took a spatial-featured data cube as input to obtain spectral features. In addition, 2D-CNN has also played an important role in HSI classification. It combines the spectral information and the spatial information by extracting the spectral-spatial features of the sample. Makantasis *et al.* [36] proposed a deep learning network for classification. In the network, R-PCA is used to reduce the dimensionality of the samples, it encoded the spatial–spectral information by a convolutional network and handed the result to the perceptron to construct the HSI classification task. Different from the proposed method, Yue *et al.* [12] directly used PCA to reduce the dimensionality and then extract the spectral-spatial features by 2D-CNN. Compared with 1D-CNN, 2D-CNN has higher accuracy in HSI classification, but it does not pay attention to the structural information of the image. Liu *et al.* [37] proposed a SS network, which adds a skip connection between encoding and decoding to improve the model.

In the second stage, researchers tried to construct a dual-channel CNN to achieve HSI classification which combines 1D-CNN and 2D-CNN. Zhang *et al.* [13] constructed a dual-branch network, it was composed of two convolutional networks, the one was 1D-CNN which was used to obtain the spectral features, and another one was 2D-CNN which was used to obtain the spatial features. The network also built pyramid structures for feature fusion. At the same time, [22] proposed a similar structure, which used 1D-CNN to process single-pixel spectral features and introduced transfer learning to deal with a few shot problems.

In the third stage, 3D-CNN extracts the spectral features and the spatial features simultaneously through the 3-D convolution kernel. The simple, intuitive, and powerful structure of 3D-CNN makes it widely used in HSI classification tasks [38], [39]. Ying *et al.* [40] proposed a 3D-CNN to extract the spectral-spatial features of the image simultaneously. Zhong *et al.* [41] proposed a model based on residual structure, named spectral-spatial residual network (SSRN), which further extracts spectral features by integrating spectral residuals and spatial residuals. To improve training efficiency, scholars further optimized the model. Zhang *et al.* [42] proposed a lightweight network that simplifies operations on the original model, and the lightweight network is also suitable for handling a few shot problems. Roy *et al.* [43] proposed a network named Hybrid spectral CNN (HybridSN), which uses 3-D convolution first and then stacks 2-D convolution. This method fully extracts the spatial spectral features of the image while avoiding the model complexity caused by full 3-D convolution. Liu *et al.* [44] proposed two principles for spectral-spatial feature extraction on HSI. Based on the two principles, they designed a central attention module and a dense connection strategy to extract more useful features. To preserve the main spectral properties, Liu *et al.* [45] proposed an unsupervised dimensionality reduction method named super pixel wise collaborative representation graph embedding (SPCRGE), the method also alleviates the problem of intraclass difference and interclass similarity.

B. Domain Adaptation

Domain adaptation aims to reduce the distribution discrepancies of different domains by feature mappings. Many researchers tended to build a model to realize subspace alignment. In 2017, Hao Lu *et al.* [46] proposed an algorithm named tensor-aligned invariant subspace learning (TAISL). The algorithm considers that the shared subspace needed to be obtained by transformation, it added a series of transformation metrics to the objective function. Qin *et al.* [47] considered that the same category maintains the same geometric

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distribution in different distributions. Therefore, to preserve geometry information, they introduced a manifold regularization into the model. Though these methods can solve the problem of distribution difference, they ignored the problem of insufficient samples in the target domain.

To solve the problem, many scholars try to make the knowledge learned from the source domain helpful for the classification task in the target domain, that is, domain adaptation. To narrow the distance between different domains, many methods of measuring the distribution distance between different domains have been proposed [48], [49] [50], [51]. In addition, methods of using domain classifiers to obtain the distribution difference measurement are proposed. These methods train feature extractors and domain discriminators in an adversarial manner. They hope that the features extracted by the feature extractor cannot be judged whether it belongs to the source domain or the target domain, and the domain discriminator can distinguish them well. Thus, the model can extract the domain-invariant features of the two domains. Saito *et al.* [52] proposed a domain adaptive method for unsupervised object detection. It consists of two domain discriminators; one is used to distinguish the global features and another one is used for the local features. Li *et al.* [34] proposed a new loss function for the discriminator and the model can match interdomain features by minimizing the loss.

C. Cross Domain HSI Classification

With the development of domain adaptation in computer vision, it has also achieved remarkable results in HSI classification. Early cross-domain hyperspectral classification methods were based on machine learning. In 2008, Rajan *et al.* [53] proposed a transfer learning framework using a binary hierarchical classifier. Sun *et al.* [54] introduced the multiple-kernel support vector machines (SVMs) into the model for HSIs classification. Persello and Bruzzone [55] proposed a kernel-based feature selection method for transfer learning, which minimizes the distance between two domains in a reproducing kernel Hilbert space. These methods have achieved good results in cross-domain hyperspectral classification, but they cannot perform well while facing the problem that there is a big difference between the source domain and the target domain.

Then, the researchers turned their focus on the methods of deep learning. Ji *et al.* [56] proposed a 3-D convolutional autoencoder (3D-CAE) to learn the spatial–spectral features of the HSIs, which are from the source domain and the target domain. To reduce the statistical and geometric shift, Zhang *et al.* [57] proposed a method of discriminative cooperative alignment based on subspace and distribution. They also introduce a data reconstruction to extract a more robust subspace. Huang *et al.* [58] proposed a method named graph embedding and distribution alignment (GEDA). The method uses the method of graph embedding to learn the local discriminant information, it minimizes the spectral difference by distribution alignment. Zhang *et al.* [59] proposed a graph information aggregation cross-domain few-shot learning (Gia-CFSL) framework, which performs graph alignment at feature and distribution levels. Zhang *et al.* [60] developed a topological structure and semantic information transfer network (TSTnet), which align topological relationships learned by the graph. Liu *et al.* [61] analyzed that different scenes and surrounding pixels will produce spectral shifts, and proposed spectral shift mitigation (SSM) to achieve better cross-scene classification performance, which includes amplitude shift mitigation and adjacency effect mitigation.

D. Few Shot Learning

Traditional deep learning methods often rely on large amounts of data. However, in reality, it is often not possible to obtain enough labeled samples. How to use little labeled samples to train a model is an important problem. To solve the problem, the researchers began to work on the few shot learning. This method can use prior knowledge to draw conclusions quickly from new tasks with limited supervised labeled data. The methods of few shot learning can be divided into three parts, model-based few shot learning, metric-based few shot learning, and optimization-based few shot learning.

The methods of model-based few shot learning update parameters quickly on a small number of samples through the design of the model structure. Cai *et al.* [62] proposed a memory matching network (MM-Net). This model encodes by a memory module, the encoded features and the corresponding category label form a "key-value pair" and store in the memory module. The query set is classified by comparing it with the stored features.

The methods of metric-based few shot learning measure the distance between the samples in the query set and the samples in the support set; the classification task is realized by the idea of the nearest neighbor (NN). In 2015, Koch *et al.* [63] used a pair of CNNs which share the same structure, the networks also share the parameters. The model discriminates whether the two images belong to the same category or not by a probability score. Li *et al.* [64] proposed a module named category traversal module (CTM). The module improves the feature extraction network by learning the intra and interclass features. Snell *et al.* [65] proposed a prototype network to realize few shot learning. The network obtains the prototype by calculating the mean of each category, it maximizes the distance between samples of different categories and minimizes the distance between samples of the same category. Pal *et al.* [66] proposed a network for HSI classification named stable prototypical network (SPN), which averages the parameters using the Monte Carlo. Li *et al.* [67] proposed a prototypical network named active-learning-based prototypical network (ALPN), which uses SS clustering to obtain the prototype.

The methods of optimization-based few shot learning realize the task of few-shot classification (FSC) by adjusting the optimization method. Finn *et al.* [68] proposed a model-agnostic meta-learning mechanism. The mechanism maximizes the sensitivity of the loss function for the new task to the parameters, so the model can achieve fast learning of the model for the new task. Rusu *et al.* [69] proposed a method named latent embedding optimization (LEO) to introduce a low-dimensional latent space and update model parameters.

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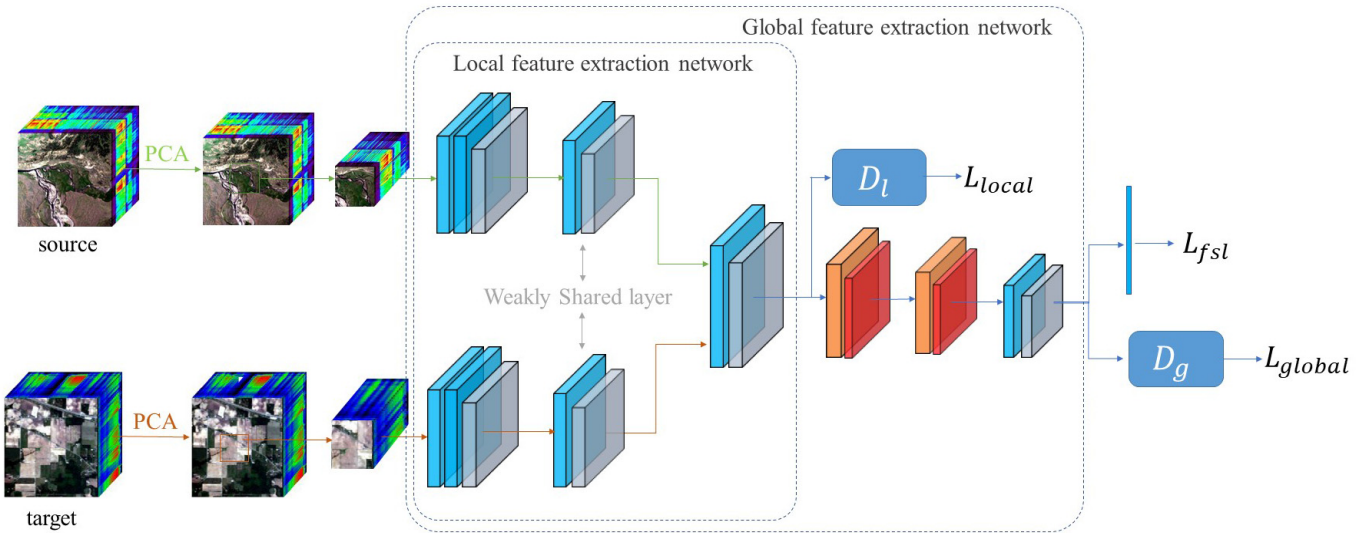


Fig. 1. Architecture of the proposed method. The module contains the weakly parameter-shared layer, the local spatial–spectral alignment module, and the global alignment module. We add the weakly parameter-shared mechanism to the low-level feature extraction network. The global feature extraction network is divided into two parts: the local feature extraction network and the rest. The features extracted by the local feature extraction network are the input of the local spatial–spectral alignment module, and the features extracted by the global feature extraction network are the input of the global alignment module.

III. Proposed Domain Adaptation Method

The architecture of the proposed method is shown in Fig. 1, which contains three main parts: weakly parameter-shared layers, local spatial–spectral alignment, and classifiers of FSL.

Given two domain datasets: a dataset with a lot of labeled samples which is called the source domain D_S and a dataset with little labeled samples which is called the target domain D_T. The two domains belong to different distributions and different category spaces. The number of the categories in the source domain is denoted as C_S. Similarly, the number of the categories in the target domain D_T can be denoted as C_T, where C_S is not necessary to be equal to C_T. At the same time, the target domain contains some categories which do not belong to the source domain. According to whether the data is labeled, we can divide the target domain data can be divided into two parts, labeled data D_l and unlabeled data D_u.

To simulate the few shot learning task, we need to select samples from the source domain and target domain to form episodic training tasks. To describe this process, we choose the source domain training task as an example. We select C categories from the dataset randomly, and extract K labeled samples for each category to form a query set, which is expressed as $Q = \{(x_i, y_i)\}_{i=1}^{C\times K}$. Then, we extract N samples from the C classes that do not belong to the query set to form a support set in which the samples have no labels, it can be denoted as $S = \{x_j\}_{j=1}^{C\times N}$. The training tasks in the target domain are similar to those in the source domain.

A. Weakly Parameter-Shared Layer

There are two common parameter-sharing mechanisms: strongly parameter-shared and weakly parameter-shared. The strongly parameter-shared mechanism forces the parameters between two models to be constant. The low-level features tend to favor domain-specific features and the high semantic features tend to favor task-specific features. Forcing the parameters of the low-level feature extraction network to be constant will ignore domain-specific features and cannot extract the most representative features. Different from the strongly parameter-shared mechanism, weakly parameter-shared mechanism allows different models have different parameters. Therefore, this method can narrow the difference between the network parameters of the source domain and the target domain. The feature networks can preserve domain-specific features and extract better representative features.

For different domains, we use different feature extraction networks to extract features. The networks share the same structure with different parameters. They both share some common features, but also have their own domain-specific features. Therefore, using the same feature extraction network is easy to over-mix the extracted features of the two domains, some features which are called domain-specific features will be ignored during the training process. To solve this problem, we introduce a weakly parameter-shared mechanism that allows extracting both domain-invariant features and domain-specific features. As shown in Fig. 1, the features obtained by the shared layers are a mixture of the features learned from two domains. Although the features we are more concerned about are the shared features of two domains, it is risky to ignore the domain-specific features completely. There is a lot of information that cannot be expressed by the shared network layer. For the input feature x, the general feature extraction network can be represented by the function $f(x) = Wx + b$. We denote the specific network of the source domain and of the target domain as F^S and F^T, respectively. Then the network can be expressed as

$$\begin{cases} f^S(x) = w^S x + b^S \\ f^T(x) = w^T x + b^T. \end{cases} \tag{1}$$

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Different domains pass through different network layers with the same structure to retain domain-specific features. After that is a strong parameter-shared layer which mainly obtains the deep feature shared by the two domains. At the same time, to balance the domain-invariant features and the domain-specific features, we divide domain-specific network F_{loc} into three parts F_{loc}^1, F_{loc}^2 and F_{loc}^3 as shown in Fig. 1, and parameter sharing constraints are added to F_{loc}^2

$$L_w = \left\| w_2^S - w_2^T \right\|_F^2 + \left\| b_2^S - b_2^T \right\|_F^2 \tag{2}$$

where, w_2^S and b_2^S represent the parameters of the F_{loc}^2 for the source domain, w_2^T and b_2^T represent the parameters of the F_{loc}^2 for the target domain. Minimizing L_w can reduce the parameter difference between the two network layers.

B. Local Spatial–Spectral Alignment

Global features mainly describe the overall properties of the entire image, including edge features and texture features of the image, which are affected by domain differences. The domain difference can be narrowed to a certain extent by aligning the global features. Local features mainly describe patches of the image and the number of local patches is large. Aligning global features alone may leave some inference errors since local domain differences are not considered. However, the local features alignment can reduce these errors by concerning on the local area. Therefore, aligning local features can further achieve the alignment of sample features between source and target domains.

Most domain adaptive methods focus on global feature alignment. Although this method achieves domain transfer learning to a certain extent, it has an obvious flaw. When the generator network can perfectly deceive the discriminator network, it means that the global marginal distributions of the source domain and the target domain are almost completely aligned. During this process, many changes have taken place in the extracted features of each area, which may cause locally extracted features to deviate from the original category and result in a decline in the classification effect. It means that forcing the global alignment of the source domain and the target domain can easily make the feature map to the wrong category, and lead to the negative migration of the model. To this end, we introduce a local domain alignment module, which consists of two classifiers, a local space-level classifier and a spectral-level classifier. The module tries to distinguish whether the sample belongs to the source domain or the target domain. By this method, the module can realize the local spatial–spectral alignment and reduce the classification error.

Compared with global features, the local domain alignment module pays more attention to local features. It contains two modules. The spatial feature extraction module is used to realize spatial alignment. The module consists of 2-D convolution layers, it mainly extracts the spatial features of samples. The classifier discriminates whether the spatial features belong to the source domain or the target domain. As shown in Fig. 1, the global feature extraction network is denoted as F, the local feature extraction network is denoted as F_{loc}, and the output of F_{loc} is the input of the local alignment module. As shown in Fig. 2, we can divide the local alignment module into two classifiers, $D_{\text{loc}}^{\text{space}}$ and $D_{\text{loc}}^{\text{spectral}}$. The loss function of the local spatial alignment module can be summarized as

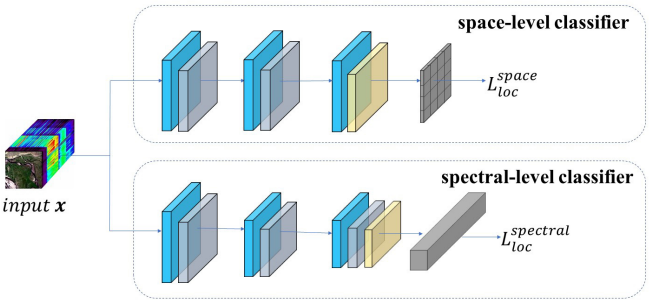


Fig. 2. Architecture of the local alignment module. The local alignment module is divided into two parts: the spatial feature extraction module and the spectral feature extraction module.

$$L_{\text{loc}_s}^{\text{space}} = \frac{1}{HW} \sum_{w=1}^{W} \sum_{h=1}^{H} \log\left(D_{\text{loc}}^{\text{space}}(F_{\text{loc}}(x_s))\right) \tag{3}$$

$$L_{\text{loc}_t}^{\text{space}} = \frac{1}{HW} \sum_{w=1}^{W} \sum_{h=1}^{H} \log\left(1 - D_{\text{loc}}^{\text{space}}(F_{\text{loc}}(x_t))\right) \tag{4}$$

where $L_{\text{loc}_s}^{\text{space}}$ represents the spatial alignment loss of the source domain, $L_{\text{loc}_t}^{\text{space}}$ represents the spatial alignment loss of the target domain, and $D_{\text{loc}}^{\text{space}}$ means the result of the spatial-level classifier. The loss is designed to align each pixel of the source domain with the target domain.

The spectral feature extraction module is used to realize the spectral alignment. The module consists of three convolution layers. It keeps the spectral dimension unchanged and mainly extracts the spectral features of the samples. The classifier discriminates whether the spectral features belong to the source domain or the target domain. The loss function of the local spectral alignment module is summarized as

$$L_{\text{loc}_s}^{\text{spectral}} = \frac{1}{C} \sum_{c=1}^{C} \log\left(D_{\text{loc}}^{\text{spectral}}(F_{\text{loc}}(x_s))\right) \tag{5}$$

$$L_{\text{loc}_t}^{\text{spectral}} = \frac{1}{C} \sum_{c=1}^{C} \log\left(1 - D_{\text{loc}}^{\text{spectral}}(F_{\text{loc}}(x_t))\right) \tag{6}$$

where $L_{\text{loc}_s}^{\text{spectral}}$ and $L_{\text{loc}_t}^{\text{spectral}}$, respectively, represent the spectral alignment loss of the source domain and the target domain, $D_{\text{loc}}^{\text{spectral}}$ means the result of the spectral-level classifier. Above all, we can get the loss of the local alignment module

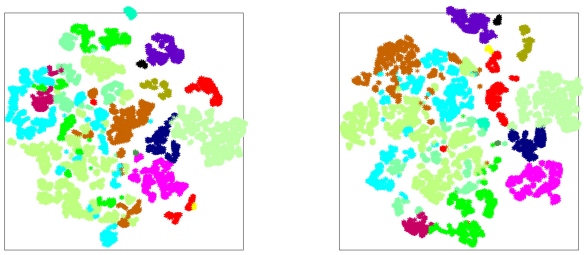
$$L_{\text{loc}} = L_{\text{loc}_s}^{\text{space}} + L_{\text{loc}_t}^{\text{space}} + L_{\text{loc}_s}^{\text{spectral}} + L_{\text{loc}_t}^{\text{spectral}}. \tag{7}$$

To realize the global alignment between two domains, we also need a global classifier. To distinguish the sample belonging to the source domain or the target domain. Thus, we can get the loss function as the following:

$$\begin{aligned} L_{\text{global}} = {} & E_{x_s \sim P_s} \log\left(D_{\text{global}}\left(F_{\text{global}}(x_s)\right)\right) \\ & + E_{x_t \sim P_t} \log\left(1 - D_{\text{global}}\left(F_{\text{global}}(x_t)\right)\right) \end{aligned} \tag{8}$$

where $D_{\text{global}}(\bullet)$ represents the output of the global classifier, respectively, $F_{\text{global}}(\bullet)$ represents the output of the global

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(a) (b)

Fig. 3. Feature visualization of the IP datasets by samples that are discriminated after feature extraction with the t-SNE algorithm. Compared to (a), (b) shows a better clustering result. It shows that the local spatial–spectral alignment module preserves the class distance and improves the classification performance.

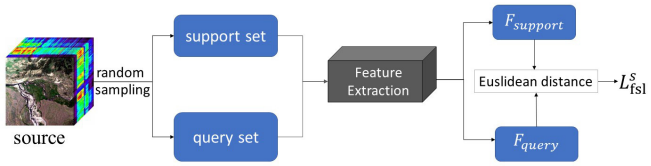


Fig. 4. Process of FSL of the source domain. The model selects K samples per class to form the support set and Q samples per class to form the query set randomly. F_{support} and F_{query} represent the features extracted by the feature extractor from the support set and the query set, respectively.

features extraction network, x_s is the sample of the source domain and x_t is the sample of the target domain. $F_{\text{global}}(\bullet)$ extract the global features.

To describe the effect of this module more vividly, Fig. 3(a) shows the t-distributed stochastic neighbor embedding (T-SNE) visualization of the features extracted from the indian pine (IP) dataset by the model with the module, and Fig. 3(b) shows it by the model without the module. From the figure, we can see that the distance between homogeneous features is closer, and the distance between heterogeneous features is farther. For example, the distance between the red dots shrinks.

C. Few Shot Learning

Few shot learning is applied to the source domain and the target domain at the same time. It transfers the knowledge learned from the source domain to the target domain. The workflow is shown in Fig. 4. For the source domain, first, we select C classes and then we select K samples from the C classes to form the support set of the training task. After that Q samples different from the K samples are selected to form the query set. To ensure that the process of few shot learning on the source and target domains is consistent, we keep the number of query sets on the source and target domains the same for each iteration. Therefore, for the target domain, we also need to select Q classes and then K samples from the labeled dataset D_l. The category distribution of the query set can be represented by the support set as follows [65]:

$$p_\theta(y = k \mid q \in Q_s) = \frac{\exp(-d(f_\theta(q), c_k))}{\sum_{k=1}^{C} \exp(-d(f_\theta(q), c_k))} \tag{9}$$

where θ denotes the parameter of the whole network, $f_\theta(\bullet)$ denotes the features extracted by the network, q represents the query sample and y is the label of q, c_k represents the embedding features of kth class, and $d(\bullet)$ is a Euclidean distance function which computes the distance between c_k and $f_\theta(\bullet)$.

According to the data and the label, we can get the loss as follows:

$$L^s_{\text{fsl}} = E_{S_s, Q_s}\left[-\sum_{(x,y)\in Q_s} \log p_\theta(y = k \mid q)\right] \tag{10}$$

$$L^t_{\text{fsl}} = E_{S_t, Q_t}\left[-\sum_{(x,y)\in Q_t} \log p_\theta(y = k \mid q)\right] \tag{11}$$

$$L_{\text{fsl}} = L^s_{\text{fsl}} + L^t_{\text{fsl}} \tag{12}$$

where S_s denotes the support set and Q_s denotes the query set of the source domain, and S_t, Q_t represents the support set and the query set of the target domain, respectively.

D. Global Information Dimensionality Reduction

Few shot learning only learns limited labeled sample information, while a large number of unlabeled sample information are ignored. Since the datasets are collected by different sensors, their spectral dimensions are different. So the dimensionality reduction is performed on the data before the feature extraction network. The method of DCFSL uses the mapping layer which only uses labeled data. To make full use of the unlabeled samples, we choose to perform PCA [70] on the entire dataset. The reduced dimension after PCA is 100. Before training, we use PCA to extract the main feature components of the source domain and the target domain to reduce dimensions. During the process, PCA takes the relationship between all samples in the dataset into account, both labeled samples and unlabeled samples. Therefore, the information of the unlabeled samples is introduced with PCA and the extracted features will be more representative.

E. Overall Objective

The proposed method utilizes weakly parameter-shared layer, local spatial–spectral alignment module and few shot learning to realize the feature alignment. The local spatial–spectral alignment module and the global alignment module are implemented based on the idea of adversarial learning. The classifiers are used to discriminate whether the spectral features belong to the source domain or the target domain. The feature extraction networks are used to confuse the source and target domain features. Therefore, we can get the adversarial loss. The adversarial loss $L_{\text{adv}}(F, D)$ is summarized as

$$L_{\text{adv}}(F, D) = L_{\text{loc}}(F_{\text{loc}}, D_{\text{loc}}) + L_{\text{global}}\left(F, D_{\text{global}}\right) \tag{13}$$

where D represents two discriminators, D_{loc} and D_{global}, F means the whole feature extraction network.

The overall objective loss function is

$$L = L_w + L_{\text{adv}}(F, D) + L_{\text{fsl}}. \tag{14}$$

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TABLE I
SAMPLE DISTRIBUTION OF THE CHIKUSEI

Class	Name	Samples
1	Water	2845
2	Bare soil (school)	2859
3	Bare soil (park)	286
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (grown)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (non-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145

The training process corresponds to solving for a target model according to the optimization

$$\min_{F} \max_{D} L_{\text{adv}}(F, D) + L_{\text{fsl}} + L_{w} \tag{15}$$

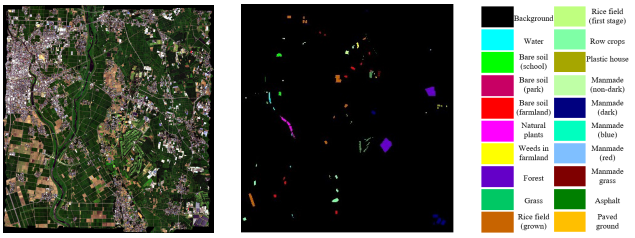
where L_{fsl} represents the loss of the few shot learning, L_w is the loss of the weakly parameter-shared layer. The discriminators minimize the above function and the generators maximize it.

IV. EXPERIMENTS

A. Dataset

To access the performance of the model, we used five different HSI datasets, including the University of Pavia (UP), the Center of Pavia (PC), Salinas, IPs, and Chikusei. Among them, the Chikusei is selected to be the source domain and the other four datasets are selected to be the target domains.

1) Source domain: The Chikusei dataset is an aerial hyperspectral dataset collected by the University of Tokyo in Chikusei City, Japan using the Headwall Hyperspec-visible and near-infrared C-Series (VNIR-C) sensor, including 128 bands ranging from 343 nanometers to 1018 nanometers. The size of the dataset is 2517*2335, the spatial resolution is 2.5 m, and there are 19 categories, including urban and rural areas. In Fig. 5, the false-color image can be seen. The figure also shows the ground-truth map. The categories of the Chikusei dataset are shown in Table I with the number of each category.
2) Target domain
   a) The UP dataset is collected by reflective optics system imaging spectrometer (ROSIS) sensors, with a total of 103 bands, its size is 610*340, and it contains nine categories. The false-color image can be seen in Fig. 6 with the ground-truth map. The categories of the UP dataset are shown in Table II with the number of each category.

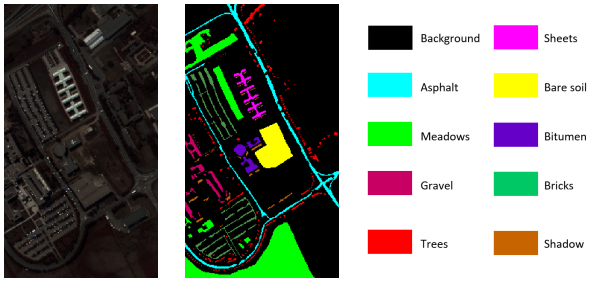


(a) (b) (c)

Fig. 5. Chikusei. (a) False-color image. (b) Ground-truth map. (c) Categories.

TABLE II
SAMPLE DISTRIBUTION OF THE PAVIA UNIVERSITY

Class	Name	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947



(a) (b) (c)

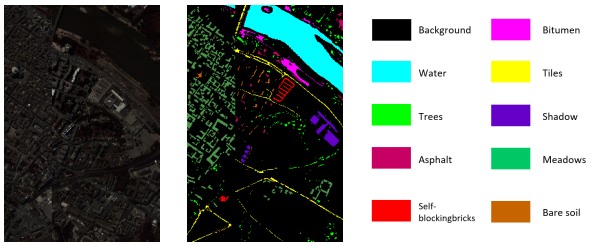
Fig. 6. UP. (a) False-color image. (b) Ground-truth map. (c) Categories.

   b) The PC dataset is collected by the ROSIS sensor. It contains 102 bands, the size is 1096*715, and contains nine feature categories, which are the same as the UP dataset. The sample categories of the PC dataset and the number of each category are shown in Table III. The false-color image can be seen in Fig. 7.
   c) The Salinas dataset was collected by airborne visible infrared imaging spectrometer (AVIRIS) sensors in the Salinas Valley, California. After removing the 20 bands that are severely affected by water vapor, the dataset has a total of 204 bands with a size of 512*217, a spatial resolution of 3.7 m, and 16 crop categories. In Fig. 8, we can see the false-color image and the ground-truth map. The categories of the Salinas dataset are shown in the Table IV with the number of each category.
   d) The IP dataset was collected by AVIRIS sensors in Indiana. There are 200 effective bands, while removed 20 bands that cannot be reflected by water. The datasets size is 145*145, and has

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TABLE III
SAMPLE DISTRIBUTION OF THE PAVIA CENTER

Class	Name	Samples
1	Water	824
2	Trees	820
3	Asphalt	816
4	Self-Blocking Bricks	808
5	Bitumen	808
6	Tiles	1260
7	Shadows	476
8	Meadows	824
9	Bare Soil	820



(a) (b) (c)

Fig. 7. PC. (a) False-color image. (b) Ground-truth map. (c) Categories.



(a) (b) (c)

Fig. 8. Salinas. (a) False-color image. (b) Ground-truth map. (c) Categories.

TABLE IV
SAMPLE DISTRIBUTION OF THE SALINAS SCENE

Class	Name	Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807

16 crop categories. The false-color image is shown in Fig. 9 with the ground-truth map. The categories distribution of the IP dataset is shown in Table V.

TABLE V
SAMPLE DISTRIBUTION OF THE INDIAN PINES

Class	Name	Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93



(a) (b) (c)

Fig. 9. IP. (a) False-color image. (b) Ground-truth map. (c) Categories.

B. Experimental Setting

To ensure the fairness of the experiment, we compared our results with some other supervised methods, such as 3-D-CNN [40], SSRN [71], deep few-shot learning (DFSL)+ NN [72], relation network (RN)-FSC [73], and DCFSL [34].

We use N_C to represent the number of categories, and N_K is used to denote the sample number of each class. To simulate the training process of few shot learning, in each episode, we need to randomly select N_C categories from the source domain and the target domain, then we select N_K pixels for each class and obtain 9×9 pixel blocks around the pixels as the space size of the input cube. We set the number of training rounds to 10 000, which is sufficient to fully train the model. The convergence speed of the model is determined by the learning rate and we set it to 0.001. As far as the few shot learning is concerned, the class number of each episode is also important. In this experiment, we set N_K equal to 5. The value of N_C is different according to different datasets. The value of N_C for the UP dataset and the PC dataset, which have nine categories, is nine. For the Salinas dataset and the IP dataset which contain 16 categories, the value of it is 16. We denote the number of query samples for each class as N_Q. In theory, in the training process, the larger the value of N_Q, the better the model simulates the classification task of the test set. But considering the computational efficiency and the limitation of training samples, according to the number of categories of the source domain Chikusei is 19, we set the N_Q value to 19. We use overall accuracy (OA), average accuracy, and kappa to evaluate all the methods.

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TABLE VI
CLASSIFICATION RESULTS (%) OF THE UP DATASET

Class	3-D-CNN	SSRN	DFSL+KNN	RN-FSC	DCFSL	OURS
1	59.82	**95.44**	87.35	68.55	84.62	84.99
2	63.05	**95.69**	95.23	89.25	89.50	85.57
3	68.91	58.36	59.98	49.81	71.32	**78.49**
4	77.31	**96.38**	77.59	92.15	88.33	83.01
5	90.77	**99.97**	99.79	99.43	99.86	99.05
6	63.40	40.84	54.79	57.99	71.71	**82.34**
7	**97.64**	80.33	48.21	70.04	83.41	91.50
8	57.27	72.47	62.77	67.37	60.98	**80.81**
9	95.57	98.74	96.99	**99.19**	96.19	91.50
OA	65.74	76.96	75.32	80.19	83.51	**85.06**
AA	73.72	81.82	75.85	77.12	89.50	**87.18**
Kappa	57.37	71.18	69.08	73.73	78.36	**80.60**

TABLE VII
CLASSIFICATION RESULTS (%) OF THE PC DATASET

Class	3-D-CNN	SSRN	DFSL+KNN	RN-FSC	DCFSL	OURS
1	98.31	**99.99**	99.94	99.64	99.86	99.88
2	82.29	**95.51**	92.92	94.48	91.17	89.01
3	86.63	81.72	72.35	**94.10**	91.78	93.81
4	78.86	74.32	63.68	93.58	92.18	**95.21**
5	87.01	**97.57**	95.47	84.25	91.22	91.79
6	83.60	85.11	89.62	89.83	**97.80**	97.58
7	82.16	**96.44**	90.03	81.44	84.61	88.59
8	96.12	**99.90**	99.68	97.43	98.65	98.56
9	99.56	98.98	**99.60**	98.53	97.61	99.25
OA	94.07	96.77	96.44	96.30	97.45	**97.66**
AA	88.28	91.88	94.97	92.59	93.88	**94.85**
Kappa	91.66	95.44	89.26	94.78	96.40	**96.69**

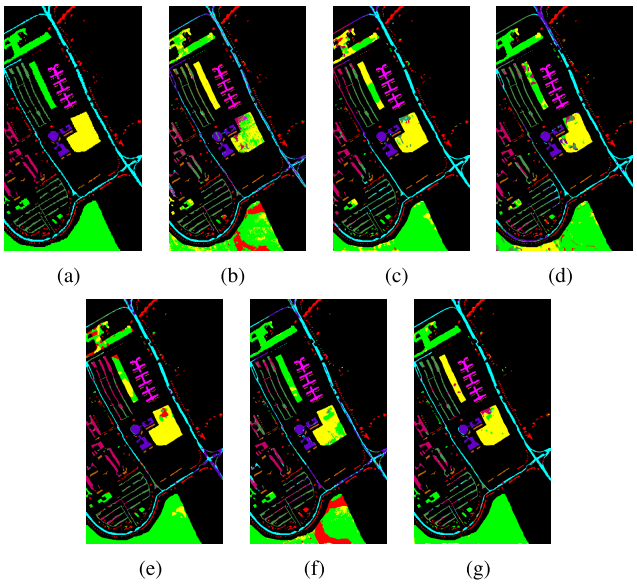


Fig. 10. UP. (a) Ground-truth map. (b)–(g) Classification results for different methods. (b) 3-D-CNN. (c) SSRN. (d) DFSL+NN. (e) RN-FSC. (f) DCFSL. (g) Our method.

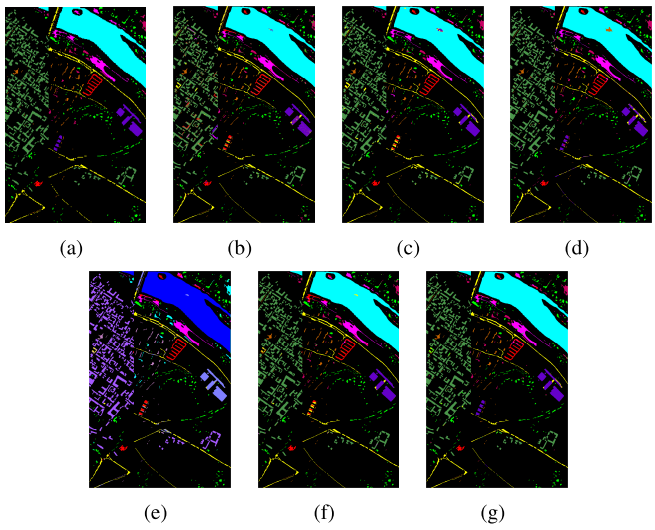


Fig. 11. PC. (a) Ground-truth map. (b)–(g) Classification results for different methods. (b) 3-D-CNN. (c) SSRN. (d) DFSL+NN. (e) RN-FSC. (f) DCFSL. (g) Our method.

C. Experimental Results

For the supervised learning methods SSRN, they can train a classifier by just a few shot samples of the target domain. So that we only need to select five samples for each class from the target domain randomly. These samples are used to train the classification model and the rest are used to test. Especially, for the few shot learning methods DFSL+ k-nearest neighbor (KNN) and RN-FSC, we choose four datasets as the source domains, not only Chikusei, but also three other datasets, such as Houston, Botswana, and Kennedy Space Center. The four datasets have a total of 61 classes. Among them, 21 classes with less than 200 samples are removed. Finally, we use the remaining 40 classes as the training set. To ensure the dimensionality is consistent, we use the method of the graph representation based band selection (GRBS) [74], which may be called as GRBS simply, to reduce the dimensionality of the dataset to 100. For the cross-domain method DCFSL, we choose five samples for each class from the source and target domains, respectively. In our proposed method, we select five samples per class to form the support set and select 200 samples to form the query set. We use PCA instead of GRBS to select bands before training. In addition, to eliminate the impact of sampling randomly, the experiments on four target datasets were taken ten times and computed the mean. The results with different methods are shown in Tables VI–IX.

3D-CNN is a 3-D CNN that is used to extract the spectral features and the spatial features simultaneously. SSRN is an end-to-end SSRN. It combines spectral information with spatial information to extract classification features and it is helpful for the network to obtain discriminative features for classification. The two methods can train the model with few labeled samples and have a good performance. DFSL+KNN introduces the meta learning method into the deep network,

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TABLE VIII
CLASSIFICATION RESULTS (%) OF THE IP DATASET

Class	3-D-CNN	SSRN	DFSL+KNN	RN-FSC	DCFSL	OURS
1	95.12	59.53	59.73	**96.34**	83.90	92.33
2	37.70	**61.69**	55.39	46.13	46.10	60.18
3	19.77	48.37	55.56	40.61	52.29	**62.56**
4	32.51	45.53	31.60	58.62	**77.58**	76.66
5	88.45	**92.95**	92.80	64.96	63.93	82.42
6	73.65	90.29	**99.09**	69.45	90.01	90.54
7	81.82	60.11	29.39	**100**	**100**	98.13
8	53.35	**99.90**	99.71	77.70	85.96	91.06
9	**100**	39.58	12.58	**100**	**100**	99.04
10	41.35	42.20	48.17	25.49	64.41	**65.61**
11	66.71	68.84	**75.93**	65.51	60.62	66.01
12	37.40	48.16	52.37	27.13	**53.47**	45.74
13	85.71	98.66	85.28	**99.75**	99.70	96.28
14	62.57	**99.98**	96.90	76.35	88.98	89.10
15	56.42	46.72	67.10	74.10	69.55	**84.70**
16	90.36	75.39	86.94	**100**	98.18	97.07
OA	54.76	64.02	66.21	58.17	66.86	**72.28**
AA	63.93	59.82	65.53	69.90	77.18	**81.09**
Kappa	48.72	64.87	62.41	52.52	62.72	**68.81**

TABLE IX
CLASSIFICATION RESULTS (%) OF THE SALINAS DATASET

Class	3-D-CNN	SSRN	DFSL+KNN	RN-FSC	DCFSL	OURS
1	95.29	**99.90**	99.85	96.47	99.08	99.42
2	97.20	99.29	99.57	99.47	**99.88**	99.65
3	91.45	**98.03**	91.12	85.05	93.75	96.35
4	97.31	92.35	97.63	98.75	**99.45**	97.95
5	91.24	**98.51**	97.33	83.45	95.45	92.32
6	98.8	**100.00**	99.99	96.73	98.70	99.45
7	**99.69**	99.67	99.31	99.61	99.64	99.04
8	66.40	74.89	81.44	72.11	75.99	**82.97**
9	96.25	98.16	99.54	88.35	99.75	**99.78**
10	70.72	92.96	**93.38**	70.53	84.45	85.67
11	93.15	94.15	89.53	90.03	93.96	**96.33**
12	99.65	96.69	96.65	93.15	99.35	**99.72**
13	92.63	98.93	98.46	98.54	97.94	**99.43**
14	93.56	90.76	91.81	96.43	97.69	**97.82**
15	68.02	**78.42**	58.49	70.18	72.08	78.23
16	81.41	99.29	**99.89**	82.39	89.63	95.22
OA	84.20	86.39	87.61	84.11	89.08	**91.62**
AA	89.56	93.24	93.38	88.83	93.55	**94.96**
Kappa	82.46	84.95	86.25	82.38	87.86	**90.68**

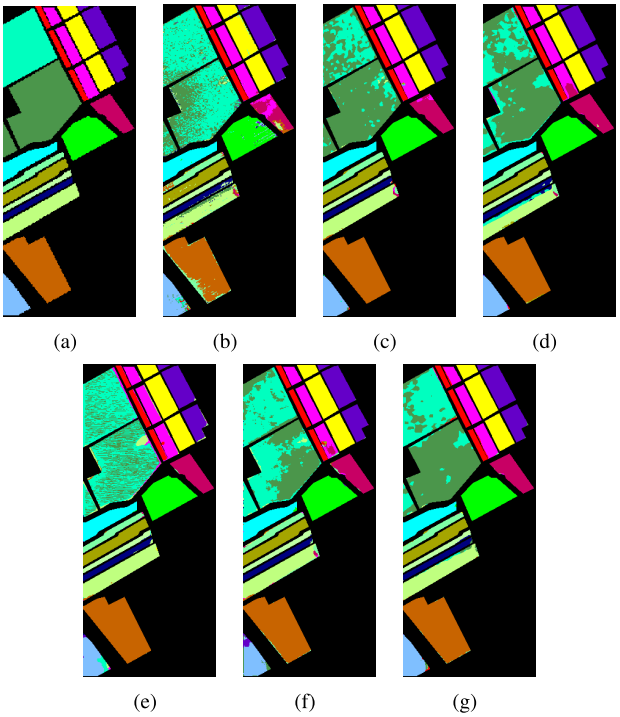


Fig. 12. Salinas. (a) Ground-truth map. (b)–(g) Classification results for different methods. (b) 3-D-CNN. (c) SSRN. (d) DFSL+NN. (e) RN-FSC. (f) DCFSL. (g) Our method.

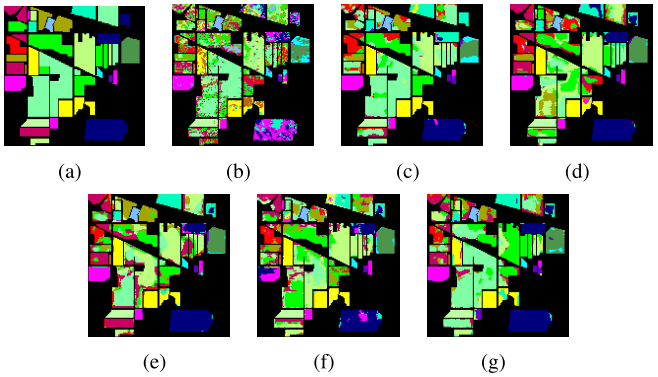


Fig. 13. IP. (a) Ground-truth map. (b)–(g) Classification results for different methods. (b) 3-D-CNN. (c) SSRN. (d) DFSL+NN. (e) RN-FSC. (f) DCFSL. (g) Our method.

because of this, the method can extract more transferable features from the source domain and the knowledge learned from these features is helpful for the target domain. Same as DFSL+KNN, RN-FSC also utilizes meta-learning to solve the problem of few shot learning. The difference between the two deep few shot methods is that RN-FSC requires fine-tuning the model in the testing process. Compared with SSRN, DFSL+KNN, and RN-FSC, DCFSL has better classification results in the target domain. It combines domain adaptation with few shot learning. This method transfers the knowledge learned from the source domain to the target domain, which can solve the problem of few shot learning to a certain extent.

By comparing our method and the above methods, we can find that our method has the best classification results in the target domain. Our method uses the information of unlabeled samples to extract more general features. It also uses a weakly parameter-shared mechanism to retain domain-specific features. To reduce classification of error, local spatial–spectral alignment is applied in the model. Compared with DCFSL, our

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TABLE X

CLASSIFICATION RESULTS (%) OF FOUR DATASET, W INDICATES WEAKLY PARAMETER-SHARED MECHANISM, L INDICATES LOCAL SPATIAL–SPECTRAL ALIGNMENT AND P INDICATES PCA

P	W	L	UP	PC	IP	SALINAS
			83.65±1.77	97.45±0.41	66.81±2.37	89.34±2.19
✓			83.96±3.74	97.49±0.22	69.27±3.29	90.35±1.83
✓	✓		84.12±2.93	97.54±0.24	70.17±4.03	90.53±1.31
✓		✓	84.07±0.86	97.49±0.18	69.45±1.19	91.17±0.76
✓	✓	✓	**85.06±1.49**	**97.66±0.27**	**72.28±2.37**	**91.62±0.71**

TABLE XI

CLASSIFICATION RESULTS (%) ON THE UP DATASET OF DIFFERENT METHODS(L INDICATES THE NUMBER OF LABELED SAMPLES PER CLASS)

	SS-CNN	DFSL+KNN	DCFSL	OURS
L1	34.72±4.08	49.86±6.36	57.55±6.41	**58.05±6.36**
L2	41.43±8.36	62.61±7.22	68.03±5.29	**71.16±4.12**
L3	48.44±3.89	66.99±3.99	74.07±4.58	**76.97±4.64**
L4	51.20±4.56	71.46±6.05	80.32±3.40	**80.37±3.58**
L5	54.48±10.57	75.32±4.51	83.51±1.06	**85.06±1.49**

TABLE XII

CLASSIFICATION RESULTS (%) ON THE PC DATASET OF DIFFERENT METHODS(L INDICATES THE NUMBER OF LABELED SAMPLES PER CLASS)

	SS-CNN	DFSL+KNN	DCFSL	OURS
L1	47.41±20.69	86.43±5.75	91.55±2.67	**91.72±2.54**
L2	73.44±4.85	92.00±2.42	95.15±1.37	**95.37±0.72**
L3	76.38±6.18	94.76±1.75	96.82±0.43	**96.87±0.66**
L4	77.12±4.22	95.58±1.39	97.00±0.55	**97.10±0.30**
L5	80.83±2.89	96.44±0.55	97.45±0.48	**97.66±0.27**

method increases OA by 1.55%, 0.21%, 5.42%, and 2.54% in four target datasets, respectively.

To further compare the above-mentioned methods, Figs. 10–13 shows the classification result maps of different datasets on different methods. We can intuitively find that compared with other methods, the result map obtained by our method is most similar to the true value map.

D. Ablation Experiment

To more accurately illustrate the effectiveness of each module, we show the performance of PCA, weakly parameter-shared layer, and local spatial–spectral alignment (P, W, and L in Table) on four target datasets. From Table X, we can find that our methods constantly perform better than the baseline. At the same time, compared to the method with PCA, the method which adds the weakly parameter-shared layer and the method which adds the local spatial–spectral alignment achieve higher accuracy; the result shows that each module is effective.

TABLE XIII

CLASSIFICATION RESULTS (%) ON THE SALINAS DATASET OF DIFFERENT METHODS(L INDICATES THE NUMBER OF LABELED SAMPLES PER CLASS)

	SS-CNN	DFSL+KNN	DCFSL	OURS
L1	55.46±4.66	70.30±4.59	72.70±3.43	**74.49±4.62**
L2	56.85±5.83	79.16±3.10	80.88±2.77	**83.46±4.42**
L3	65.86±2.41	83.23±3.84	85.50±1.69	**86.26±2.93**
L4	70.95±3.96	85.31±2.84	88.49±0.66	**89.55±2.46**
L5	71.76±1.68	87.61±1.85	89.08±0.75	**91.62±0.71**

TABLE XIV

CLASSIFICATION RESULTS (%) ON THE IP DATASET OF DIFFERENT METHODS(L INDICATES THE NUMBER OF LABELED SAMPLES PER CLASS)

	SS-CNN	DFSL+KNN	DCFSL	OURS
L1	32.18±6.46	50.86±3.96	43.00±3.87	**52.09±4.37**
L2	43.43±5.66	57.98±3.71	57.89±3.49	**58.53±5.72**
L3	43.67±3.26	60.81±4.08	58.64±1.94	**63.85±3.10**
L4	48.03±1.11	64.28±1.32	63.06±2.03	**70.73±3.31**
L5	51.39±1.46	66.21±1.71	66.86±1.45	**72.28±2.37**

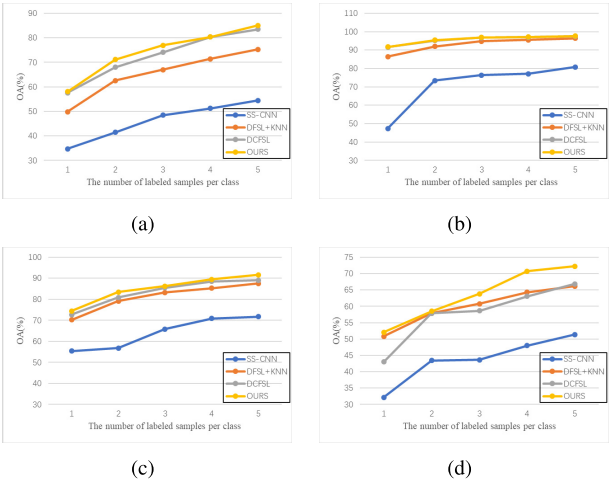


Fig. 14. Classification results with different number of labeled samples on four target datasets. (a) UP. (b) PC. (c) Salinas. (d) IP.

We take 1, 2, 3, 4, and 5 labeled samples per class for analysis from four target datasets and the experimental results are illustrated in Tables XI–XIV. We show the impact of different numbers of labeled samples per class on the classification results. With the increase of the number of labeled samples, the classification accuracy also improves. To better illustrate the superiority of our method, we also show the comparison with some other methods of few shot learning (SS-CNN [37], DFSL-KNN [72], and DCFSL [34]). SS-CNN uses encoder and decoder to form a SS CNN to extract spectral-spatial features directly. Compared with SS-CNN, DFSL-KNN introduces a meta-learning method to extract more transferable features. DCFSL combines domain

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TABLE XV
CLASSIFICATION RESULTS (%) ON FOUR DATASETS OF DIFFERENT SOURCE DOMAIN(L INDICATES THE NUMBER OF LABELED SAMPLES PER CLASS)

Target	Source	L1	L2	L3	L4	L5
UP	Chikusei	58.05±6.36	71.16±4.12	76.97±4.64	80.37±3.58	85.06±1.49
	Houston	57.69±10.86	68.56±4.01	70.16±3.93	73.81±5.54	79.84±6.19
PC	Chikusei	91.72±2.54	95.37±0.72	96.87±0.66	97.10±0.30	97.66±0.27
	Houston	85.91±7.51	94.09±1.52	96.20±0.86	97.03±0.33	97.56±0.31
SALINAS	Chikusei	76.97±4.64	83.46±4.12	86.26±4.64	89.55±3.58	91.62±0.71
	Houston	76.88±4.79	79.43±3.99	87.17±3.77	87.25±2.81	89.64±2.68
IP	Chikusei	52.09±4.37	58.53±5.72	63.85±3.10	70.73±3.31	72.28±2.37
	Houston	40.21±6.36	46.47±7.59	55.66±3.21	62.71±2.38	65.54±2.94

TABLE XVI
TIME COMPARISON (S) OF EACH DATASET BETWEEN DIFFERENT METHODS

		3-D-CNN	SS-CNN	SSRN	DFSL+KNN	RN-FSC	DCFSL	OURS
UP	Training time	53.86	1212.41	66.56	435.03	310.28	1240.94	1089.51
	Testing time	6.23	3.85	30.32	5.00	46.29	5.11	4.93
PC	Training time	54.39	2037.19	66.49	594.99	391.50	560.80	2066.37
	Testing time	21.50	14.66	106.37	17.43	72.97	17.84	50.04
SALINAS	Training time	63.03	674.55	83.58	946.63	495.20	1267.57	1520.51
	Testing time	8.36	2.86	41.43	6.54	55.53	7.45	5.92
IP	Training time	62.35	109.78	145.28	706.71	450.34	1255.45	1590.72
	Testing time	1.50	0.50	7.6	1.28	42.68	1.39	1.26

adaptation with meta-learning to transfer the knowledge learned from the source domain to the target domain efficiently. Compared with the above methods, our method utilizes both labeled and unlabeled information, and further aligns the source and target domains. From Fig. 14, we can find that our method has better results intuitively.

To illustrate the importance of source domain dataset selection, we compare the classification accuracy of Chikusei and Houston. As can be seen from Table XV, Chikusei performs better than Houston. Comparing the two datasets, it is found that the Chikusei dataset contains more samples and more categories that are consistent with the target domain, that is, the smaller the difference between the source domain and the target domain, the better the knowledge transfer effect.

In Table XVI, we compare the training and testing time of different methods, we can find that our method requires the most training time. However, the classification accuracy of our method is the highest with similar inference time.

V. CONCLUSION

This article proposes a new method to solve the cross-domain few shot problem of HSI classification. It utilizes a meta-learning mechanism to transfer the knowledge of the source domain to the target domain. We balance domain-invariant features and domain-specific features by a weakly parameter-shared mechanism. Besides this, the local spatial–spectral alignment module is introduced to reduce the classification error. To utilize both labeled data and unlabeled data, we employ PCA on both source and target domain data in the preprocessing stage. This method cannot make full use of a large amount of unlabeled data information. In the future research direction, self-supervised methods can be considered to fully learn interdomain information. Experimental results on four different datasets show that our method has a better performance in HSI classification based on cross-domain few shot learning.

REFERENCES

[1] D. Tuia, C. Persello, and L. Bruzzone, "Domain adaptation for the classification of remote sensing data: An overview of recent advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 41–57, Jun. 2016.

[2] Q. Du and C.-I. Chang, "A linear constrained distance-based discriminant analysis for hyperspectral image classification," *Pattern Recognit.*, vol. 34, no. 2, pp. 361–373, 2001.

[3] L. Samaniego, A. Bardossy, and K. Schulz, "Supervised classification of remotely sensed imagery using a modified k-NN technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2112–2125, Jul. 2008.

[4] L. Sun, Z. Wu, J. Liu, L. Xiao, and Z. Wei, "Supervised spectral–spatial hyperspectral image classification with weighted Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1490–1503, Mar. 2015.

[5] Y. Zhong, A. Ma, and L. Zhang, "An adaptive memetic fuzzy clustering algorithm with spatial information for remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1235–1248, Apr. 2014.

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[6] Z. Lin, Y. Chen, X. Zhao, and G. Wang, "Spectral–spatial classification of hyperspectral image using autoencoders," in *Proc. 9th Int. Conf. Inf., Commun. Signal Process.*, Dec. 2013, pp. 1–5.

[7] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[8] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[9] H. Zhang and Y. Li, "Spectral–spatial classification of hyperspectral imagery based on deep convolutional network," in *Proc. Int. Conf. Orange Technol. (ICOT)*, Dec. 2016, pp. 44–47.

[10] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[11] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[12] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, Jun. 2015.

[13] H. Zhang, Y. Li, Y. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery using a dual-channel convolutional neural network," *Remote Sens. Lett.*, vol. 8, no. 5, pp. 438–447, 2017.

[14] B. Liu, X. Yu, P. Zhang, X. Tan, A. Yu, and Z. Xue, "A semi-supervised convolutional neural network for hyperspectral image classification," *Remote Sens. Lett.*, vol. 8, no. 9, pp. 839–848, Sep. 2017.

[15] S. Rajan, J. Ghosh, and M. M. Crawford, "An active learning approach to hyperspectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 4, pp. 1231–1242, Apr. 2008.

[16] D. Tuia, F. Ratle, F. Pacifici, M. F. Kanevski, and W. J. Emery, "Active learning methods for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 7, pp. 2218–2232, Jul. 2009.

[17] M. M. Crawford, D. Tuia, and H. L. Yang, "Active learning: Any value for classification of remotely sensed data?" *Proc. IEEE*, vol. 101, no. 3, pp. 593–608, Mar. 2013.

[18] Z. Sun, C. Wang, H. Wang, and J. Li, "Learn multiple-kernel SVMs for domain adaptation in hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1224–1228, Sep. 2013.

[19] J. Xia, N. Yokoya, and A. Iwasaki, "Ensemble of transfer component analysis for domain adaptation in hyperspectral remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 4762–4765.

[20] G. Matasci, M. Volpi, M. Kanevski, L. Bruzzone, and D. Tuia, "Semi-supervised transfer component analysis for domain adaptation in remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3550–3564, Jul. 2015.

[21] J. Blitzer, R. McDonald, and F. Pereira, "Domain adaptation with structural correspondence learning," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2006, pp. 120–128.

[22] Q. Gu, Z. Li, and J. Han, "Joint feature selection and subspace learning," in *Proc. 22nd Int. Joint Conf. Artif. Intell.*, 2011, pp. 1294–1299.

[23] J. Li, J. Zhao, and K. Lu, "Joint feature selection and structure preservation for domain adaptation," in *Proc. IJCAI*, 2016, pp. 1697–1703.

[24] H. Sun, S. Liu, S. Zhou, and H. Zou, "Unsupervised cross-view semantic transfer for remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 13–17, Jan. 2016.

[25] C. Luo and L. Ma, "Manifold regularized distribution adaptation for classification of remote sensing images," *IEEE Access*, vol. 6, pp. 4697–4708, 2018.

[26] T. Liu, X. Zhang, and Y. Gu, "Unsupervised cross-temporal classification of hyperspectral images with multiple geodesic flow kernel learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9688–9701, Dec. 2019.

[27] Y. Zhu *et al.*, "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, Apr. 2020.

[28] Y. Qu, R. K. Baghbaderani, W. Li, L. Gao, Y. Zhang, and H. Qi, "Physically constrained transfer learning through shared abundance space for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10455–10472, Dec. 2021.

[29] Y. Mansour, M. Mohri, and A. Rostamizadeh, "Domain adaptation with multiple sources," in *Proc. Conf. Neural Inf. Process. Syst.*, 2008, pp. 1041–1048.

[30] S. Zhao *et al.*, "Multi-source distilling domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 7, Apr. 2020, pp. 12975–12983.

[31] X. Liang, Y. Zhang, and J. Zhang, "Attention multisource fusion-based deep few-shot learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8773–8788, 2021.

[32] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.

[33] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[34] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[35] H. Zhang and Y. Li, "Spectral–spatial classification of hyperspectral imagery based on deep convolutional network," in *Proc. Int. Conf. Orange Technol. (ICOT)*, Dec. 2016, pp. 44–47.

[36] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[37] B. Liu, X. Yu, P. Zhang, X. Tan, A. Yu, and Z. Xue, "A semi-supervised convolutional neural network for hyperspectral image classification," *Remote Sens. Lett.*, vol. 8, no. 9, pp. 839–848, Sep. 2017.

[38] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, 2017.

[39] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.

[40] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[41] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Oct. 2017.

[42] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[43] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3D–2D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[44] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 10, 2022, doi: 10.1109/TNNLS.2022.3155114.

[45] H. Liu, W. Li, X.-G. Xia, M. Zhang, and R. Tao, "Superpixelwise collaborative-representation graph embedding for unsupervised dimension reduction in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4684–4698, 2021.

[46] H. Lu *et al.*, "When unsupervised domain adaptation meets tensor representations," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 599–608.

[47] Y. Qin, L. Bruzzone, and B. Li, "Tensor alignment based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9290–9307, Nov. 2019.

[48] M. Long, Y. Cao, J. Wang, and M. Jordan, "Learning transferable features with deep adaptation networks," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 97–105.

[49] M. Long, H. Zhu, J. Wang, and M. I. Jordan, "Unsupervised domain adaptation with residual transfer networks," 2016, *arXiv:1602.04433*.

[50] K. Saito, K. Watanabe, Y. Ushiku, and T. Harada, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3723–3732.

[51] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7167–7176.

[52] K. Saito, Y. Ushiku, T. Harada, and K. Saenko, "Strong-weak distribution alignment for adaptive object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6956–6965.

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[53] S. Rajan, J. Ghosh, and M. M. Crawford, "Exploiting class hierarchies for knowledge transfer in hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3408–3417, Nov. 2006.

[54] Z. Sun, C. Wang, H. Wang, and J. Li, "Learn multiple-kernel SVMs for domain adaptation in hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1224–1228, Sep. 2013.

[55] C. Persello and L. Bruzzone, "Kernel-based domain-invariant feature selection in hyperspectral images for transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2615–2626, May 2016.

[56] J. Ji, S. Mei, J. Hou, X. Li, and Q. Du, "Learning sensor-specific features for hyperspectral images via 3-dimensional convolutional autoencoder," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 1820–1823.

[57] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 9646–9660, Nov. 2021.

[58] Y. Huang, J. Peng, Y. Ning, W. Sun, and Q. Du, "Graph embedding and distribution alignment for domain adaptation in hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7654–7666, 2021.

[59] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[60] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 16, 2021, doi: 10.1109/TNNLS.2021.3109872.

[61] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Spectral shift mitigation for cross-scene hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6624–6638, 2021.

[62] Q. Cai, Y. Pan, T. Yao, C. Yan, and T. Mei, "Memory matching networks for one-shot image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4080–4088.

[63] G. Koch *et al.*, "Siamese neural networks for one-shot image recognition," in *Proc. ICML Deep Learn. Workshop*, vol. 2. Lille, France, 2015, pp. 1–30.

[64] H. Li, D. Eigen, S. Dodge, M. Zeiler, and X. Wang, "Finding task-relevant features for few-shot learning by category traversal," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1–10.

[65] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4080–4090.

[66] D. Pal, V. Bundele, B. Banerjee, and Y. Jeppu, "SPN: Stable prototypical network for few-shot learning-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[67] X. Li, Z. Cao, L. Zhao, and J. Jiang, "ALPN: Active-learning-based prototypical network for few-shot hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[68] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1126–1135.

[69] A. A. Rusu *et al.*, "Meta-learning with latent embedding optimization," 2018, *arXiv:1807.05960*.

[70] I. T. Jolliffe, "Principal component analysis," *J. Marketing Res.*, vol. 87, p. 513, Aug. 2002.

[71] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[72] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Oct. 2018.

[73] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020.

[74] K. Sun *et al.*, "A robust and efficient band selection method using graph representation for hyperspectral imagery," *Int. J. Remote Sens.*, vol. 37, no. 20, pp. 4874–4889, Oct. 2016.



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